

Linear Algebra I — Week 2

Matrices over $\mathbb{C}$, Conjugate Transpose, Abstract Spaces & Subspace Tests

Name: _____ Date: _____ All work **by hand** — no calculator, no software.

Key Facts

Complex numbers. $i^2 = -1$; $\overline{a+bi} = a-bi$; $z\bar{z} = |z|^2 = a^2+b^2$; $\frac{z}{w} = \frac{z\bar{w}}{|w|^2}$. Also $\overline{z+w} = \bar{z}+\bar{w}$, $\overline{zw} = \bar{z}\bar{w}$, and $z = \bar{z} \iff z \in \mathbb{R}$.

Matrices over $\mathbb{C}$. $M_{m \times n}(\mathbb{C})$ is the set of $m \times n$ matrices with complex entries. Addition and scalar multiplication are entrywise; $(AB)_{jk} = \sum_l A_{jl}B_{lk}$. Multiplication is associative but *not* commutative.

Conjugate transpose. $(A^*)_{jk} = \overline{A_{kj}}$. Rules: $(A+B)^* = A^*+B^*$, $(\lambda A)^* = \bar{\lambda}A^*$, $(AB)^* = B^*A^*$, $(A^*)^* = A$. A is **Hermitian** if $A^* = A$; **skew-Hermitian** if $A^* = -A$.

Vector space over $\mathbb{F}$. A set V with $+$ and scalar multiplication satisfying: commutativity and associativity of $+$; a zero vector $\mathbf{0}$; additive inverses; $1v = v$; $(\lambda\mu)v = \lambda(\mu v)$; $\lambda(u+v) = \lambda u + \lambda v$; $(\lambda + \mu)v = \lambda v + \mu v$. Standard examples: $\mathbb{F}^n$, $M_{m \times n}(\mathbb{F})$, $P_n(\mathbb{F})$ (polynomials of degree $\leq n$), $\mathcal{F}(S, \mathbb{F})$ (functions $S \rightarrow \mathbb{F}$, pointwise operations).

Subspace test. $W \subseteq V$ is a subspace iff (i) $\mathbf{0} \in W$, (ii) $u, v \in W \Rightarrow u+v \in W$, (iii) $\lambda \in \mathbb{F}, v \in W \Rightarrow \lambda v \in W$. To disprove: exhibit one failure. The field matters: a set can be a subspace over $\mathbb{R}$ but not over $\mathbb{C}$.

Worked Examples

Example 1 · Complex arithmetic

Compute $(2 - i)(1 + 3i)$ and $\frac{5 + i}{2 - i}$.

Solution.

- $(2 - i)(1 + 3i) = 2 + 6i - i - 3i^2 = 5 + 5i$.
- $\frac{5 + i}{2 - i} = \frac{(5 + i)(2 + i)}{(2 - i)(2 + i)} = \frac{10 + 5i + 2i + i^2}{5} = \frac{9 + 7i}{5}$.

Answer: $5 + 5i$ and $\frac{9}{5} + \frac{7}{5}i$

Example 2 · Product over $\mathbb{C}$

Compute $\begin{pmatrix} i & 1 \\ 0 & 2 \end{pmatrix} \begin{pmatrix} 1 & i \\ i & 0 \end{pmatrix}$.

Solution.

- Row 1: $(i \cdot 1 + 1 \cdot i, i \cdot i + 1 \cdot 0) = (2i, -1)$.
- Row 2: $(0 \cdot 1 + 2 \cdot i, 0 \cdot i + 2 \cdot 0) = (2i, 0)$.

Answer: $\begin{pmatrix} 2i & -1 \\ 2i & 0 \end{pmatrix}$

Example 3 · Conjugate transpose

Find M^* for $M = \begin{pmatrix} 2 - i & 3 \\ i & 1 + 2i \end{pmatrix}$.

Solution.

- Transpose: $\begin{pmatrix} 2 - i & i \\ 3 & 1 + 2i \end{pmatrix}$.
- Conjugate every entry.

Answer: $M^* = \begin{pmatrix} 2 + i & -i \\ 3 & 1 - 2i \end{pmatrix}$

Example 4 · Hermitian test

Is $N = \begin{pmatrix} 1 & 2 + i \\ 2 - i & 3 \end{pmatrix}$ Hermitian?

Solution.

- N^* : transpose swaps $2 + i$ and $2 - i$; conjugating gives $\overline{2 - i} = 2 + i$ in position $(1, 2)$ and $\overline{2 + i} = 2 - i$ in position $(2, 1)$.
- Diagonal entries 1, 3 are real and unchanged.

Answer: Yes, $N^* = N$.

Example 5 · Proof: $(AB)^* = B^*A^*$

Prove $(AB)^* = B^*A^*$ for $A \in M_{m \times n}(\mathbb{C})$, $B \in M_{n \times p}(\mathbb{C})$.

Solution.

1. Entry (j, k) of the left side: $\overline{(AB)_{kj}} = \overline{\sum_l A_{kl} B_{lj}} = \sum_l \overline{A_{kl}} \overline{B_{lj}}$ (conjugation respects sums and products).
2. Entry (j, k) of the right side: $\sum_l (B^*)_{jl} (A^*)_{lk} = \sum_l \overline{B_{lj}} \overline{A_{kl}}$.
3. The two sums agree term by term (multiplication in $\mathbb{C}$ commutes).

Answer: $(AB)^* = B^*A^*$.

Example 6 · Axiom failure

On $\mathbb{R}^2$ with usual $\oplus$ and $\lambda \odot (a, b) = (\lambda a, b)$: is it a vector space?

Solution.

1. Test $(\lambda + \mu) \odot v = \lambda \odot v \oplus \mu \odot v$ with $v = (a, b)$.
2. Left: $((\lambda + \mu)a, b)$. Right: $(\lambda a + \mu a, b + b) = ((\lambda + \mu)a, 2b)$.
3. With $b \neq 0$ these differ.

Answer: No: distributivity over scalar addition fails.

Example 7 · Subspace test

Show $W = \{p \in P_2(\mathbb{R}) : p(0) = 0\}$ is a subspace.

Solution.

1. $\mathbf{0}(0) = 0$, so $\mathbf{0} \in W$.
2. $p(0) = q(0) = 0 \Rightarrow (p + q)(0) = 0$.
3. $p(0) = 0 \Rightarrow (\lambda p)(0) = \lambda \cdot 0 = 0$.

Answer: W is a subspace.

Example 8 · Non-subspace

Is $D = \{A \in M_{2 \times 2}(\mathbb{R}) : \det A = 0\}$ a subspace?

Solution.

1. $A = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$ and $B = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}$ both have $\det = 0$, so lie in D .
2. $A + B = I$, and $\det I = 1 \neq 0$.

Answer: No: not closed under addition.

Practice Problems

Part A — Complex Arithmetic and Matrices over $\mathbb{C}$

1. [Foundational]
 Compute $(3 + 2i)(1 - 4i)$ and write it in the form $a + bi$.

2. [Foundational]
 Compute $\frac{2 + i}{1 - 3i}$ and write it in the form $a + bi$.

3. [Foundational]
 Let $A = \begin{pmatrix} 1 & i \\ 2 - i & 3 \end{pmatrix}$ and $B = \begin{pmatrix} i & 0 \\ 1 & 1 + i \end{pmatrix}$. Compute $A + B$ and $2iA$.

4. [Intermediate]
 With A and B as in Problem 3, compute AB .

5. [Foundational]
 Find the conjugate transpose C^* of $C = \begin{pmatrix} 1 + i & 2 \\ 3i & 4 - i \\ 0 & -2i \end{pmatrix}$, and state its size.

6.

[Intermediate]

Decide whether each matrix is Hermitian ($M^* = M$), skew-Hermitian ($M^* = -M$), or neither: $H = \begin{pmatrix} 2 & 1-i \\ 1+i & 5 \end{pmatrix}$, $K = \begin{pmatrix} i & 1 \\ -1 & 2i \end{pmatrix}$.

7.

[Intermediate]

Find real numbers x, y such that $M = \begin{pmatrix} 2 & x+iy \\ 3-2i & 5 \end{pmatrix}$ is Hermitian.

8.

[Challenge]

Let $A = \begin{pmatrix} 1 & i \\ 2 & 1-i \end{pmatrix}$. Compute $\text{tr}(A^*A)$ without computing all of A^*A , and explain why the result is always a non-negative real number.

Part B — Matrix Algebra over $\mathbb{C}$

9.

[Foundational]

Let $A = \begin{pmatrix} 1 & i \\ 0 & 1 \end{pmatrix}$. Compute A^2 and A^3 , then state A^n .

10.

[Intermediate]

Let $A = \begin{pmatrix} 0 & i \\ 1 & 0 \end{pmatrix}$ and $B = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$. Compute AB , BA and $AB - BA$.

11.

[Intermediate]

Let $A = \begin{pmatrix} 1 & i \\ 0 & 1 \end{pmatrix}$ and $B = \begin{pmatrix} 1 & 0 \\ i & 1 \end{pmatrix}$. Verify $(AB)^* = B^*A^*$ by computing both sides.

12.

[Challenge]

Let A be an $n \times n$ Hermitian matrix. Prove that (a) every diagonal entry of A is real, and (b) iA is skew-Hermitian.

Part C — Abstract Vector Spaces

13.

[Intermediate]

On $\mathbb{R}^2$ define $(a, b) \oplus (c, d) = (a + c, b + d)$ and $\lambda \odot (a, b) = (\lambda a, 0)$. Is $(\mathbb{R}^2, \oplus, \odot)$ a vector space over $\mathbb{R}$? Justify.

14.

[Challenge]

Let $V = \mathbb{R}_{>0}$ with $x \oplus y = xy$ and $\lambda \odot x = x^\lambda$. (Assume it is a real vector space.) Find the zero vector, the additive inverse of 5, and compute $(3 \odot 2) \oplus ((-1) \odot 5)$.

15.

[Challenge]

Using only the vector space axioms, prove that $0 \cdot v = \mathbf{0}$ for every vector v (here 0 is the scalar zero and $\mathbf{0}$ the zero vector).

16.

[Foundational]

In $P_2(\mathbb{R})$ let $p(x) = 1 + x - x^2$ and $q(x) = 2 - x + x^2$. Compute $3p - 2q$.

17.

[Foundational]

In $\mathcal{F}(\mathbb{R}, \mathbb{R})$ let $f(x) = x^2$ and $g(x) = 3x + 1$. Write the function $2f - g$ and evaluate it at $x = 2$.

18.

[Intermediate]

In $\mathbb{C}^2$ over $\mathbb{C}$, compute $(1 + i)(2 - i, 3i) + (-i)(1, 1 + i)$.

Part D — Subspace Tests

19.

[Foundational]

Show that $W = \{(x, y, z) \in \mathbb{R}^3 : x + 2y - z = 0\}$ is a subspace of $\mathbb{R}^3$.

20.

[Foundational]

Show that $W = \{(x, y) \in \mathbb{R}^2 : xy = 0\}$ is *not* a subspace of $\mathbb{R}^2$.

21.

[Challenge]

Let $W = \{A \in M_{2 \times 2}(\mathbb{C}) : A^* = A\}$ (Hermitian matrices). Is W a subspace of $M_{2 \times 2}(\mathbb{C})$ over $\mathbb{C}$? Is it a subspace over $\mathbb{R}$? Justify both.

22.

[Intermediate]

In $P_3(\mathbb{R})$ let $W_1 = \{p : p(2) = 0\}$ and $W_2 = \{p : p(2) = 1\}$. Decide which is a subspace.

23.

[Intermediate]

Show that the set E of even functions $f \in \mathcal{F}(\mathbb{R}, \mathbb{R})$, i.e. $f(-x) = f(x)$ for all x , is a subspace.

24.

[Challenge]

In $\mathbb{R}^2$ let $U = \{(x, 0)\}$ and $W = \{(0, y)\}$. Show that $U \cap W$ and $U + W$ are subspaces, but $U \cup W$ is not.